Supplementary Material: Directional Prompting for Gaze-Referential Inference in Large Multimodal Models

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This document provides the items deferred from the main paper: construction details for CG-GRI, prompt templates, and an analysis of `Gemini-3.1-pro`'s sensitivity to UniGaze-predicted gaze arrows.

1 Construction of CG-GRI

CG-GRI (N=483) is derived from VideoAttentionTarget [1] (N=200) and GazeFollow [2] (N=283). Neither source provides GRI-ready candidate lists, so we construct the multiple-choice options and targets as follows. We refer to the two subsets as VAT-GRI and GF-GRI; they are reporting slices of one CG-GRI track rather than separate evaluation tracks.

VideoAttentionTarget (VAT-GRI). VideoAttentionTarget consists of video clips with dense frame-level gaze-target annotations. Because consecutive frames are visually similar, we sample by gaze sequence or clip and retain at most one frame per clip. Automatic screening uses YOLOv10L [3]: the observer head box must be valid; detections overlapping the observer box are removed; the annotated gaze point must fall within at least one remaining box; and at least two candidate boxes must remain. Candidate descriptions are drafted by an LMM and human-edited so that instances of the same class remain visually distinguishable. Every retained instance is human-annotated for observer identity, candidate list, and the target indicated by the original VAT gaze point; ambiguous images are discarded. Ground-truth arrow origins are also human-annotated eye points, while endpoints remain the original VAT gaze points. This process yields 200 VAT-GRI instances.

GazeFollow (GF-GRI). GazeFollow provides web images with gaze-region annotations but no gazed-object labels. We randomly sample 1,000 images and retain those in which the observer's eyes and gaze direction can be judged. Images with occluded or unclear eyes are discarded. For every retained image, candidate objects are drafted using YOLO and human-edited, and the gazed target is human-annotated using the original GazeFollow heatmap. Ground-truth arrows use the original eye annotation as the origin and the original gaze point as the endpoint. This process yields 283 GF-GRI instances.

Shared protocol. The target observer is marked by a red unfilled bounding-box outline in every CG-GRI condition, including the RGB baseline. The box disambiguates the observer and selects whose gaze is predicted, but it is not used as the arrow origin. Among detected faces overlapping the box, we retain the prediction whose projected origin is closest to the reference head. The question lists the human-annotated candidate descriptions and asks which candidate the marked observer is looking at.

2 Prompting Templates

Plain multiple-choice prompt. This prompt is used for all image conditions unless the evaluated method explicitly changes the instructions. On CG-GRI, "the person" is replaced by "the person in the red bounding box."

```
What is the person looking at in the image?
You must choose exactly one option from the following options:
{choices}
Please answer in the following JSON format:
{"answer": "<your chosen option>",
"reasoning": "<your reasoning>"}.
```

Textual-arrow prompt.

```
What is the person looking at in the image?
The target person's gaze ray originates from pixel ({x1}, {y1}) and ends at pixel
({x2}, {y2}).
The origin (0, 0) is at the top-left corner of the image.
You must choose exactly one option from the following options:
{choices}
Please answer in the following JSON format:
{"answer": "<your chosen option>",
"reasoning": "<your reasoning>"}.
```

LMM-generated-arrow prompt. The first stage receives no answer choices and returns a gaze ray that is rendered on the image before the plain multiple-choice prompt is applied.

```
Identify the target person's gaze direction in this image.
Do not name an object.
Return exactly one JSON object using a normalized 0-1000 image coordinate system:
{"start": [x, y], "end": [x, y]}.
The start must be near the target person's visible eyes, and the end must lie in the
gaze direction.
```

The JSON `answer` field is scored by case-insensitive exact match against the ground-truth label.

3 Sensitivity of Gemini-3.1-pro to UniGaze-Predicted Gaze Arrows

The main paper reports a significant accuracy decrease for `Gemini-3.1-pro` in the 3-candidate stratum, where RGB+A(UniGaze) underperforms RGB. Motivated by this result, we analyze its responses over the complete CG-GRI dataset.

Table 1. Model outcomes on deduplicated unions of CG-GRI instances where RGB+A(UniGaze) benefits or harms at least one model.

Diagnostic subset and outcome	`Gemini-3.1-pro`	`Qwen3.5-27B`
Benefit union: RGB+A(UniGaze) correct (N=90)	**87/90 (96.7%)**	71/90 (78.9%)
Harm union: RGB+A(UniGaze) wrong (N=83)	**70/83 (84.3%)**	48/83 (57.8%)

We compare it with `Qwen3.5-27B` to examine whether Gemini is more responsive to the UniGaze-predicted visual arrow. We use Qwen as the comparison model because it is evaluated on the same instances with identical UniGaze-predicted arrows and achieves the same overall RGB+A(UniGaze) accuracy as Gemini (77.2%), while not exhibiting Gemini's decrease in the 3-candidate stratum.

Deduplicated union analysis. We construct two outcome-selected diagnostic subsets over the complete CG-GRI dataset. The *benefit union* contains each unique instance on which RGB+A(UniGaze) corrects an incorrect RGB answer for at least one model. This union contains 90 instances. The *harm union* similarly contains each unique instance on which RGB+A(UniGaze) overturns a correct RGB answer for at least one model, yielding 83 instances. Table 1 evaluates both models on the same instances within each union.

Interpretation. When the UniGaze-predicted visual arrow benefits at least one model, Gemini ends with the correct answer more often than Qwen (96.7% vs. 78.9%). Conversely, when the arrow harms at least one model, Gemini also ends with the wrong answer more often (84.3% vs. 57.8%). This bidirectional pattern suggests that Gemini responds more strongly to the predicted visual arrow. Such cue responsiveness provides a plausible explanation for its decrease in the 3-candidate stratum: an inaccurate predicted arrow can more readily override an otherwise correct RGB decision.

References

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